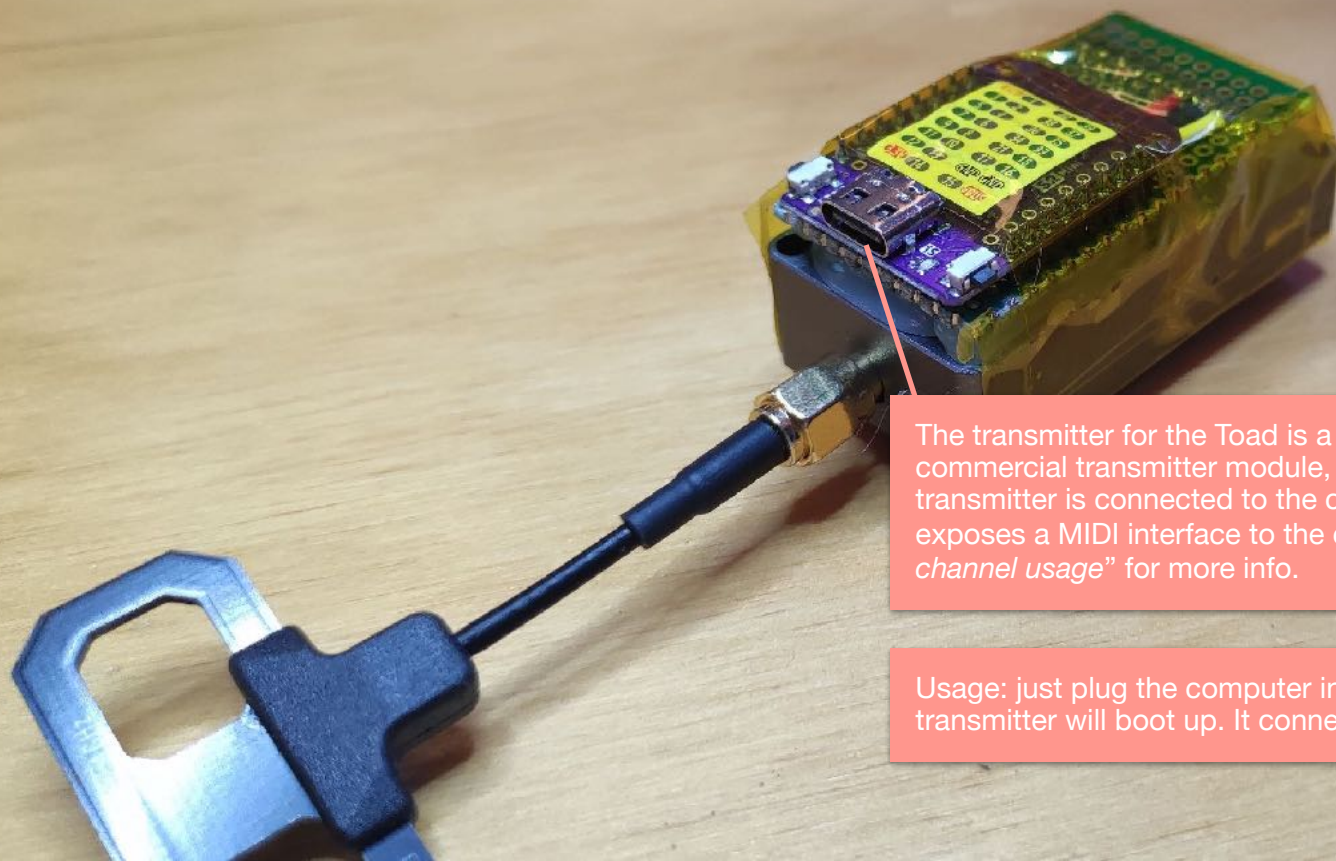




Jaha Koo - Toad Documentation

Adriaan Wormgoor
V0.8 - May 1, 2024

Transmitter: MIDI connection



The transmitter for the Toad is a microcontroller strapped to a commercial transmitter module, which it interfaces with. The transmitter is connected to the computer with a usb-c cable. This exposes a MIDI interface to the computer. See section “*MIDI channel usage*” for more info.

Usage: just plug the computer in to the usb-c port and the transmitter will boot up. It connects automatically to the Toad

Toad hardware

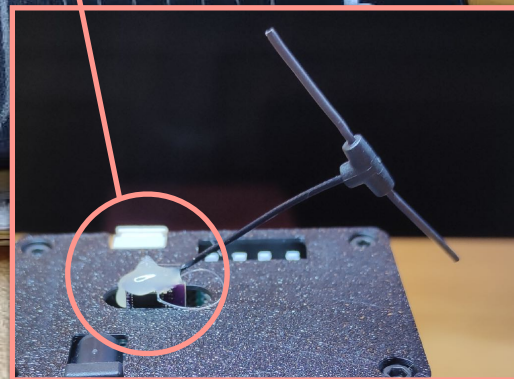
Connectors for the two eyes.
Bottom is left, top is right

USB-C connector to microcontroller
(powers receiver and eye LEDs)

The antenna of the receiver has been hot glued to keep it attached. It is flexible, but obviously please take care not to punish it too much.

12v Barrel jack (center positive)
(powers the Dynamixel motors)

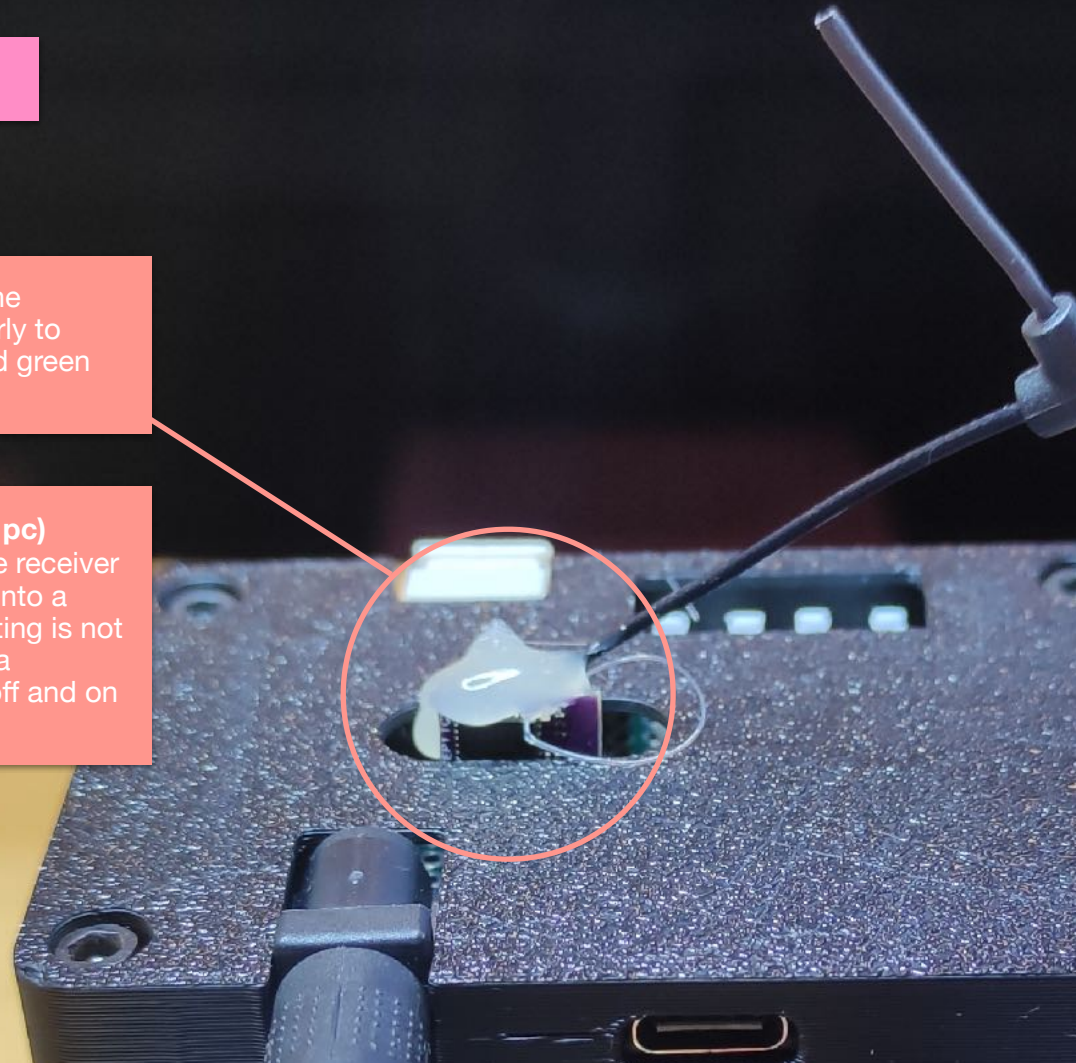
Connector to the Dynamixel motors



Toad hardware - receiver

When the microcontroller is switched on (usb-c port), the receiver will be powered. Its green LED will flash regularly to indicate it is looking for a connection, and become solid green when connected.

PLEASE always have the transmitter module (at the pc) **powered first**. This ensures a speedy connection. If the receiver hasn't made a connection within 60 seconds, it will go into a maintenance mode (rapid flashing), after which connecting is not possible. You will have to power cycle the Toad before a connection is possible again (ie: turn the battery pack off and on again).



MIDI channel usage

- When plugged in to the computer (using the usb-c port on the transmitter), the computer should detect a new MIDI device. The transmitter does not need to be switched on for this.
- All MIDI commands should be Control Change (CC) commands, over MIDI Channel 1.

Left motor - range 62->0 is backward speed, range 64->127 is forward speed. Value 63 = stopped	CC 0, range [0-127]**
Right motor	CC 1, range [0-127]**
Left eye	CC 2, range [0-127]
Right eye	CC 3, range [0-127]

** Range 62->0 = backward speed. Range 64->127 = forward speed. Value 63 = stopped